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Practical11a.c
/mnt/c/WINDOWS/system32/os/practical

```
1
2 #include <stdio.h>
3 #include <pthread.h>
4
5 long counter = 0;
6
7 void *increment(void *arg) {
8     for (int i = 0; i < 1000000; i++)
9         counter++;
10
11     return NULL;
12 }
13
14 int main() {
15     pthread_t t1, t2;
16
17     pthread_create(&t1, NULL, increment, NULL);
18     pthread_create(&t2, NULL, increment, NULL);
19
20     pthread_join(t1, NULL);
21     pthread_join(t2, NULL);
22
23     printf("Final counter: %ld\n", counter);
24
25     return 0;
26 }
```

```
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ gcc Practical11a.c -o Practical11a
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ./Practical11
-bash: ./Practical11: No such file or directory
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ./Practical11a
Final counter: 1309834
```

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Practical11b.c
/mnt/c/WINDOWS/system32/os/practical

```
1
2 #include <stdio.h>
3 #include <pthread.h>
4
5 long counter = 0;
6 pthread_mutex_t lock = PTHREAD_MUTEX_INITIALIZER;
7
8 void *increment(void *arg) {
9     for (int i = 0; i < 1000000; i++) {
10         pthread_mutex_lock(&lock);
11         counter++;
12         pthread_mutex_unlock(&lock);
13     }
14
15     return NULL;
16 }
17
18 int main() {
19     pthread_t t1, t2;
20
21     pthread_create(&t1, NULL, increment, NULL);
22     pthread_create(&t2, NULL, increment, NULL);
23
24     pthread_join(t1, NULL);
25     pthread_join(t2, NULL);
26
27     printf("Final counter: %ld\n", counter);
28
29     pthread_mutex_destroy(&lock);
30
31     return 0;
32 }
```

```
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ gcc Practical11b.c -o Practical11b
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ./Practical11b
Final counter: 2000000
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ gedit Practical11a.c
```